



NVIDIA Releases Vera Rubin DSX AI Factory Reference Design and Omniverse DSX Digital Twin Blueprint With Broad Industry Support

News Summary:

- The new NVIDIA Vera Rubin DSX AI Factory reference design provides a guide for building codesigned AI infrastructure that delivers maximum token per watt and accelerated time to first production.
- The NVIDIA Omniverse DSX Blueprint, now generally available with the NVIDIA Vera Rubin AI Factory reference design, powers digital twins for large-scale AI factory design and simulation.
- Industry leaders Cadence, Dassault Systèmes, Eaton, Jacobs, NScale, Phaidra, Procore Technologies, PTC, Schneider Electric, Siemens, Switch, Trane Technologies and Vertiv are contributing to the DSX architecture and blueprint by integrating platforms, providing SimReady assets and connecting software to help design, build and optimize AI factories.
- Energy leaders Emerald AI, GE Vernova, Hitachi and Siemens Energy are using the NVIDIA DSX reference architecture to unlock grid capacity, delivering the power needed to build and connect new AI factories.

GTC--NVIDIA today announced the NVIDIA Vera Rubin DSX AI Factory reference design, a guide for building codesigned AI infrastructure, and general availability of the [NVIDIA Omniverse DSX Blueprint](#), fully compatible with NVIDIA Vera Rubin DSX, to enable physically accurate AI factory [digital twins](#) for large-scale design, buildout and operations.

Industry leaders [Cadence](#), [Dassault Systèmes](#), [Eaton](#), [Jacobs](#), [Nscale](#), [Phaidra](#), [Procore](#), [PTC](#), [Schneider Electric](#), [Siemens](#), [Switch](#), [Trane Technologies](#) and [Vertiv](#) are contributing to the reference design and blueprint to help plan, build and operate these massive [AI factory](#) buildouts.

"In the age of AI, intelligence tokens are the new currency, and AI factories are the infrastructure that generates them," said Jensen Huang, founder and CEO of NVIDIA. "With the NVIDIA Vera Rubin DSX AI Factory reference design and Omniverse DSX Blueprint, we are providing the foundation to build the world's most productive AI factories, accelerating time to first revenue and maximizing scale and energy efficiency."

Building AI Factories That Maximize Every Watt

Building large-scale AI factories to meet the rising demand for training and inference is complex, requiring precise coordination across infrastructure, power, cooling, networking, software and compute.

The NVIDIA Vera Rubin DSX AI Factory reference design outlines how to design, build and operate the entire AI factory infrastructure stack, spanning compute, [NVIDIA Spectrum-X™ Ethernet](#) networking and storage, for repeatable, scalable and optimal cluster performance. Documentation within the reference design also equips industry partners with best practices to design, build and operate power, cooling and control systems, enabling seamless hardware-software integration and scalable deployment.

The Vera Rubin DSX software stack is open, modular and composable, connecting cluster hardware with power and cooling to maximize AI tokens per watt of available energy. Its flexible design lets AI factory builders and data center providers deploy as many components as they need.

Rubin DSX offers a collection of software libraries for partners to build on:

- **DSX Max-Q** helps AI factories and ecosystem partners maximize computing output and token performance per watt on NVIDIA systems within a fixed power budget.
- **DSX Flex** connects AI factories to power-grid services, enabling them to dynamically adjust power use and orchestrate demand with hybrid onsite generation to save energy and maintain grid stability.
- **DSX Exchange** enables scalable and secure integration of compute, network, energy, power and cooling plant signals between IT, operational technology and operations agents.
- **DSX Sim** models validate AI factories as high-fidelity digital twins, using the [NVIDIA DSX Air](#) platform to model GPUs, networking and partner infrastructure, and DSX SimReady connects detailed 3D geometry, logistics and system behavior -- accelerating time to first revenue and ensuring day-one performance.

Accelerating AI Factory Design and Simulation

Even with a detailed architecture, designing, building and operating large-scale AI factories can be difficult. Traditional design methods are limited in their ability to model entire systems, be flexible and efficient with power use, and validate

designs before construction begins.

The NVIDIA Omniverse DSX Blueprint provides an open, comprehensive framework for designing and operating large-scale AI factories. Now generally available on build.nvidia.com and fully compatible with the Vera Rubin DSX AI Factory reference design, Omniverse DSX allows developers to build physically accurate digital twins of their AI factories, simulate operations in real time and optimize performance before construction or deployment begins.

Omniverse DSX unifies power, cooling, networking and operations in one environment to accelerate time to revenue and AI efficiency. Using [NVIDIA Omniverse™](#) libraries, companies can simulate layouts, power topologies, thermal behavior and operational policies -- and evaluate hardware or workload changes without disrupting production.

Industry Leaders Embrace New Reference Design and Blueprint

Reliable, scalable power and cooling are the backbone of every AI factory, enabling intelligent systems to adapt in real time to changing compute demands while maximizing efficiency and uptime. A broad ecosystem of partners, from energy to software leaders, are embracing DSX to transform the full AI factory lifecycle into a seamlessly optimized product-level system, codesigned for resilience, precision and efficiency.

Dassault Systèmes is integrating the new reference design and blueprint into its Model Based Systems Engineering platform, powered by CATIA software, building the Virtual Twin of AI Factory to accelerate time to first revenue while improving reliability and efficiency, while Schneider Electric is [integrating its ETAP platform](#) to help users simulate and optimize power distribution systems.

[Cadence](#) is integrating simulation-ready ([SimReady](#)) models of the [NVIDIA GB300 NVL72](#) system into its Reality Data Center Digital Twin Platform to simulate thermal and fluid data to optimize AI factory design and operations, and collaborating to model NVIDIA Vera Rubin systems. [Siemens](#) is developing a framework that balances high-density compute with power, cooling and automation for AI infrastructure.

Jacobs has developed its new [Data Center Digital Twin solution](#) using the Omniverse DSX Blueprint to offer digital twins for builders and operators to optimize AI factories from planning and design through delivery and operations. [PTC](#) is integrating the blueprint into its [Windchill](#) product lifecycle management solution for DSX Accelerator, connecting engineering and product design data with high-fidelity, real-time simulation, and managing bills of materials across partners and suppliers. [Procore](#) is integrating NVIDIA Omniverse libraries and the DSX Blueprint into its Procore Platform to create a continuous digital thread across the entire construction lifecycle.

Switch [is building its EVO AI Factories and LDC EVO operating system](#) with the Omniverse DSX Blueprint for real-time telemetry ingestion, continuously updated digital twins built to Rubin DSX reference design specifications, and automated optimization of power, cooling and workloads. [Nscale](#) and Caterpillar are bringing DSX Vera Rubin reference designs to life in West Virginia on a multi-gigawatt site, one of the largest AI factories in the world.

CoreWeave is using NVIDIA DSX Air to build and test digital twins of AI factories in the cloud, achieving shortened validation time by running operational rehearsals well ahead of physical delivery.

NVIDIA's partner ecosystem provides SimReady assets of equipment and hardware, allowing AI factory operators to simulate, validate and refine designs before implementation. [Eaton](#), [Schneider Electric](#), Siemens, Trane Technologies and Vertiv are providing SimReady assets of their generators, electrical equipment and cooling systems to DSX design and engineering companies to validate and simulate entire designs before building in the physical world.

[Vertiv](#) is also using the Omniverse DSX Blueprint to build Vertiv OneCore Rubin DSX, a prefabricated, converged data center infrastructure solution designed to accelerate AI factory deployment and AI output per watt. Trane Technologies is using the blueprint to optimize thermal management for gigawatt-scale AI factories, reducing cooling plant power use and improving efficiency.

Phaidra [has integrated DSX Max-Q](#) into its new self-learning AI agent to deliver about 10% more compute by reducing cooling spikes while maintaining safety and freeing up power for revenue-generating token production.

Global Energy Leaders Modernize World's Power Grids With Omniverse DSX Blueprint

Energy is now the biggest bottleneck for AI infrastructure buildouts, with over \$300 billion in equipment backlogs and more than 200 gigawatts of projects waiting in U.S. interconnection queues.

To address this, NVIDIA is working with leading energy providers to unlock faster access to power and strengthen grid stability:

- **Emerald AI** is [integrating DSX Flex](#) with its Conductor platform to help AI factories manage power in real time -- turning demand up or down on command and coordinating flexible load with new dedicated generation -- giving utilities greater confidence to approve larger, faster grid connections through reliable, software-based load control.
- **GE Vernova** [is extending digital twin capabilities across the power stack](#) from grid to AI factories -- aligning with NVIDIA DSX reference architecture to unify power and compute modeling for faster, more accurate and predictable infrastructure deployment for large-scale AI factories.

- **Hitachi** is partnering with NVIDIA to accelerate grid planning and deliver efficient, reliable power for gigawatt-scale AI factories, combining physical AI, power systems and automation expertise with advanced computing platforms.
- **Siemens Energy** is using NVIDIA RAPIDS™ libraries, the NVIDIA Metropolis platform and the NVIDIA Isaac Sim™ framework in its Noedra digital twin platform to monitor grid health in real time, helping predict risks before failure and reduce unplanned outages.

Watch the [GTC keynote](#) from Huang and explore [sessions](#).

About NVIDIA

[NVIDIA](#) (NASDAQ: NVDA) is the world leader in AI and accelerated computing.

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